

DePeak

A simple, step-by-step guide - no experience needed

DePeak takes a load profile (how much heating power is used, hour by hour, over a year) and makes it more realistic. When you simply add up several buildings' profiles, the peak looks bigger than it really is - because not every building hits its own peak at exactly the same moment. DePeak fixes this, without changing the total amount of energy used.

This guide walks through every screen, step by step.

Step 1 - Choose an approach

DePeak offers two different ways to do this. Pick the one that fits your data.

Option A: Scaling approach (“Winter”)

1. Choose approach

☒ Scaling approach (Winter)
☐ Shift approach

Suitable for: space heating or total heating demand profiles.

The diversity factor is calculated according to [Winter's approach](#) - only the formula comes from that paper. How the resulting peak reduction is then spread across nearby time steps is our own implementation, not part of Winter's original method.

This approach shrinks the highest points (“peaks”) of your combined profile by a calculated amount, called the **diversity factor**. Think of it like this: if 50 people each own a kettle, they will almost never all boil water at the exact same second - so the total power needed is a bit less than 50 kettles at once. The diversity factor is that “a bit less”, as one single number. Whatever energy is removed from the peaks is spread out over other times, so the total amount of energy used over the whole year stays exactly the same.

Use this for: heating demand profiles (space heating, or heating combined with hot water use). It is not yet confirmed to work correctly for cooling, electricity, or hot-water-only profiles.

Option B: Shift approach

1. Choose approach

☐ Scaling approach (Winter)
☒ Shift approach

Suitable for: any load profile type

Each building profile is randomly shifted within a time window. The window grows with the number of buildings, up to the maximum you set below. This reduces the overlap of individual peaks and therefore lowers the peak of the aggregated profile.

This approach works completely differently: instead of calculating a diversity factor, it takes each building's own profile and nudges it slightly in time - a few minutes earlier or later, picked at random. This mimics the fact that real people don't all turn their heating up at the

exact same minute. Because the buildings are now slightly “out of sync” with each other, their combined peak becomes a little lower - all on its own, without telling the tool how much reduction to aim for.

Use this for: any type of profile, in principle - this approach isn't tied to heating specifically.

Step 2 - Settings

The options you see here change depending on which approach you picked in Step 1.

If you chose Scaling (Winter)

2. Settings

Lower limit of diversity factor

0.45

Controls how strongly the load peaks are reduced. Lower values result in stronger peak reduction, while higher values result in weaker peak reduction.

Time resolution

hourly

1

Lower limit of diversity factor - a number between 0 and 1. Move it left for a stronger effect (bigger buildings peaks get cut down more), or right for a gentler effect. If you're not sure, the default shown is a reasonable starting point.

2

Time resolution - how finely your data is split up: every hour, every 15 minutes, or every minute. This should match the resolution of the file you're about to upload.

If you chose Shift

2. Settings

Time resolution

hourly



Maximum time window (h)

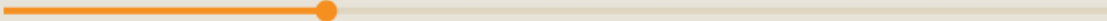


1.00



Saturation speed (τ)

80



Lower values spread the peaks over a wider time range sooner, while higher values keep them closer together for longer. With $\tau=80$, the time-shift range reaches about 50% of its maximum at 55 buildings and 90% at 184 buildings.

1

Time resolution - same idea as above: hourly, 15-minute, or 1-minute data.

2

Maximum time window - the biggest possible nudge, in hours. A value of 1.00 means a building's profile might shift by up to half an hour earlier or half an hour later (one hour total range).

3

Saturation speed - controls how quickly the shifting “warms up” as you add more buildings. You don't need to change this unless you know you want to - the explanation text underneath updates automatically to tell you what it means for your case.

Step 3 - Upload your file(s)

There are two ways to give DePeak your data. If you picked the Scaling approach, you can either upload individual profiles or one aggregated - they work equally well, it's just a matter of what your data looks like. If you picked the Shift approach, you'll always have to upload individual profiles for each building.

Option A: One combined file

Use this if you already have everything combined into one file - the totals of several buildings added together. Please type the number of buildings used further down.

3. Upload profiles

Input type

☒ Aggregated profile ☐ Individual profiles

An already aggregated profile (one file).

Aggregated profile (.csv or .xlsx)

 testpr...ldings.xlsx 0.5MB +

Sheet

Load profiles

Column number with load values Row where the time series starts End row (optional)

4 32 0

Preview of the uploaded file - the highlighted cells below show exactly what will be read as the time series.

	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10	Column 11	Column 12
30	None	None	None	None	None	None	None	None	None	None	None	None
31	None	Time	Total heat c	Space heating (l	Domestic h	Heat (ener	Total coolir	Space cool	Process co	Cooling (ene	Plug loads (l	E
32	None	Mon, 01.01	1600.40000	1543.200000	57.200000	1411.70000	0	0	0	0	0	C
33	None	Mon, 01.01	1631.20000	1597.500000	33.700000	1585.90000	0	0	0	0	0	C
34	None	Mon, 01.01	1488.60000	1470.700000	17.900000	1609.40000	0	0	0	0	0	C
35	None	Mon, 01.01	1584	1590.900000	8.100000	1687.90000	0	0	0	0	0	C

Values that will be read: 8760

Number of buildings in profile

50

Calculate

1

Sheet - if your file is an Excel file with more than one tab, choose the correct tab here. (CSV files skip this - they don't have tabs.)

2

Column number - look at the table further down on the screen: it shows you your file exactly as it is, with columns numbered 1, 2, 3, and so on. Find the column with your actual load numbers in it, and type that column's number here.

3

Row where the time series starts - in that same table, find the row number where the REAL numbers begin (not a title or header row above them). Type that row number here. Don't worry about getting this exactly right - if you pick a row that only has text in it, DePeak automatically looks further down for you until it finds real numbers.

4

End row (optional) - leave this at 0 to read all the way to the end of the file. Only change this if your file contains more data than you want to use (for example, several years' worth, but you only want one year).

5

The orange cells - this is the most important part of the whole screen. These highlighted cells show you EXACTLY which numbers will actually be used, based on what you typed above. Always check this before continuing.

6

"Values that will be read" - a plain count of how many numbers were found. A full year of hourly data should show 8760 (or 8784 in a leap year). If you see a much smaller or much bigger number than you expect, something is probably set wrong above - go back and check.

7

Number of buildings in profile - since this is ONE combined file, DePeak can't count the buildings by itself. Tell it here how many buildings were added together to make this one file.

Option B: One file per building

Use this if you have several separate files, one per building. DePeak adds them together itself - and since it can see exactly how many files you uploaded, it already knows the number of buildings without you having to type it in.

3. Upload profiles

Individual building profiles (one file per building).

Please make sure all uploaded files share the same structure (same column position, same starting row) - e.g. all exported from the same program. Files with a different layout may be read incorrectly.

Individual profiles (.csv or .xlsx)

testprofile1.xlsx
363.1KB

testprofile2.xlsx
363.1KB

testprofile3.xlsx
363.1KB

+

Sheet

Load profiles

Column number with load values

Row where the time series starts

End row (optional)

4 32 0

Preview of the uploaded file - the highlighted cells below show exactly what will be read as the time series.

	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10	Column 11	Column 12
30	None	None	None	None	None	None	None	None	None	None	None	None
31	None	Time	Total heat c	Space heating	Domestic h	Total coolin	Space cooli	Process coo	Electricity c	Plug loads (l	E-mobility (l	Air temperat
32	None	Mon, 01.01	1.800000	1.700000	0.100000	0	0	0	0	0	0	-1.800000
33	None	Mon, 01.01	1.900000	1.800000	0.100000	0	0	0	0	0	0	-1.900000
34	None	Mon, 01.01	1.600000	1.600000	0	0	0	0	0	0	0	-1.800000
35	None	Mon, 01.01	1.700000	1.700000	0	0	0	0	0	0	0	-1.600000

Values that will be read: 8760

These settings (column/row) are applied to ALL uploaded files below. Check that every file was read correctly - especially if they come from different sources/programs:

File	Values read
testprofile1.xlsx	8760
testprofile2.xlsx	8760
testprofile3.xlsx	8760

Calculate

1

Important warning: all of your uploaded files need to look the same way inside (same column, same starting row) - for example, all exported from the same program. DePeak uses the SAME column/row settings for every file you upload, so a file that's built differently may be read incorrectly without any error message.

2

Your uploaded files - each one shown as a small tile. You can add more with the "+" button, or remove one by clicking the small X.

3

Column, 4 Row, and 5 End row - exactly the same idea as for a single aggregated file (see the previous page). The preview table below is based on your FIRST uploaded file - but the same settings get applied to every file in the list.

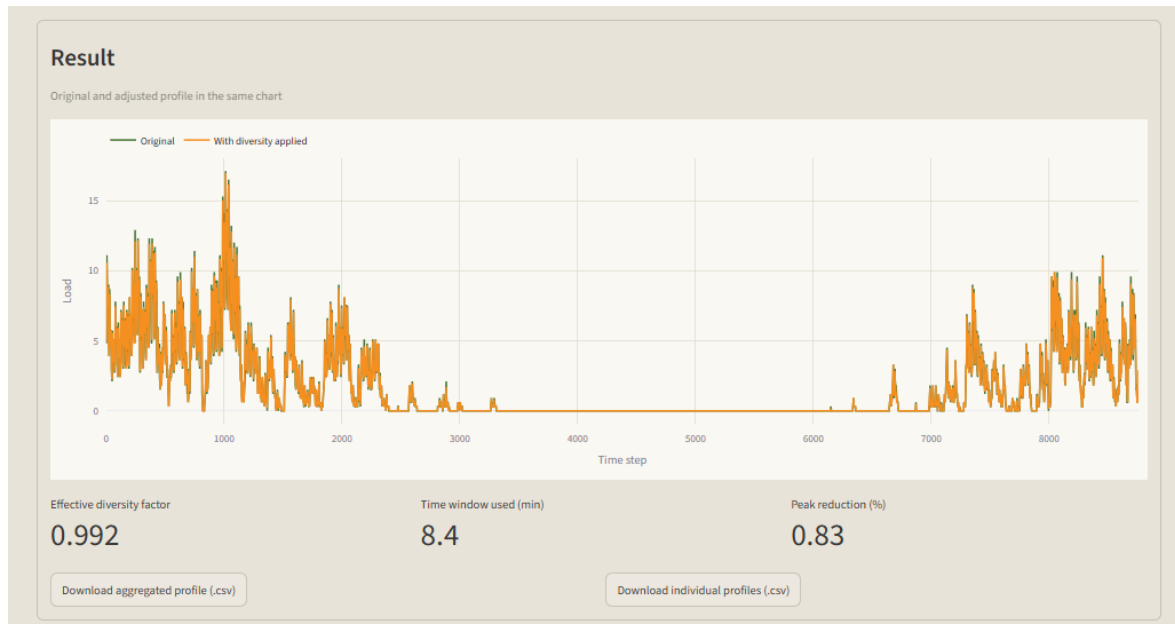
6

The check table at the bottom - since the same settings are used for every file, this table lets you double-check that each individual file was actually read correctly. If one file shows a very different number of "Values read" than the others, that file probably has a different layout and needs to be fixed or uploaded separately.

Step 4 - Click Calculate, and read your result

Once everything above looks right, press the orange **Calculate** button. Here's what you'll see, for both approaches:

Result of the Shift approach

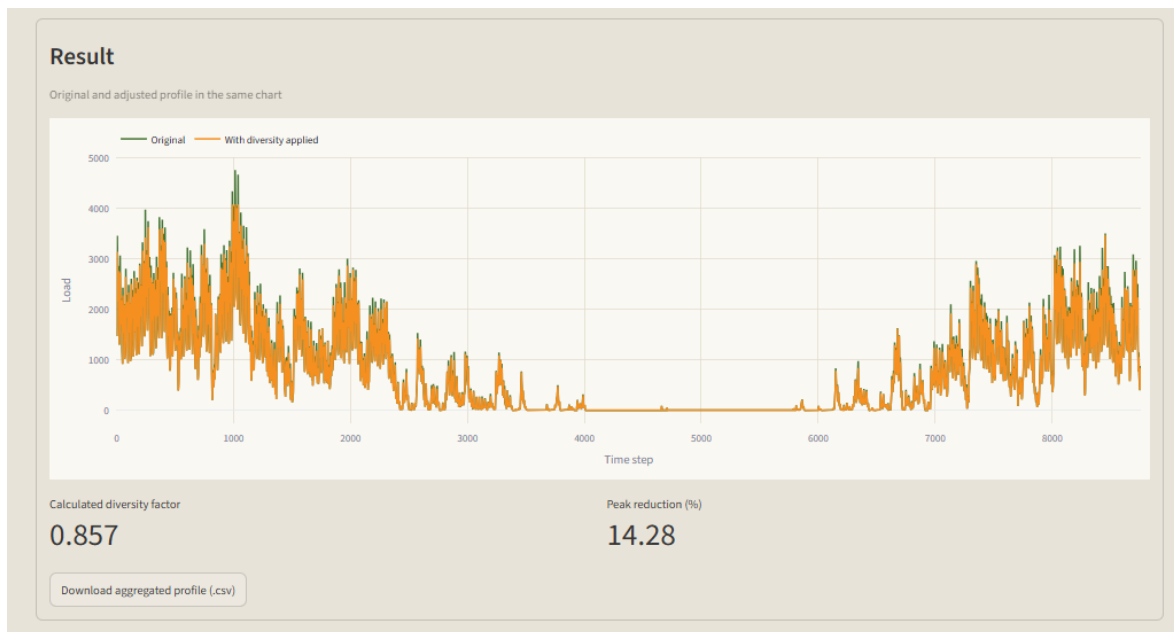


The chart shows your original combined profile (green) and the adjusted one (orange) on top of each other, across the whole year.

Effective diversity factor - how much the peak actually went down, shown as a ratio (a value close to 1.0 means barely any change). **Time window used** - the actual shifting window that was used, in minutes. **Peak reduction (%)** - the same effect, but shown as a simple percentage drop in the highest point.

Two download buttons appear: one for the combined result, and one for the individual, per-building profiles after shifting.

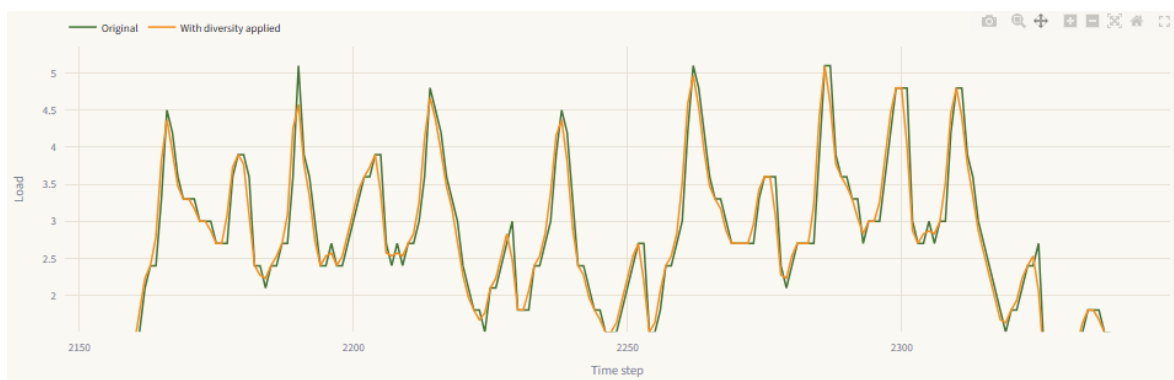
Result of the Scaling approach



Same chart idea as above. **Calculated diversity factor** is the number worked out from your settings in Step 2. **Peak reduction (%)** shows how much lower the highest point became. Since this approach doesn't shift individual buildings, only one download button appears, for the combined result.

That's it - that's the whole tool. If anything still isn't clear, check the Documentation and Contact sections at the bottom of the DePeak page.

Tip: you can zoom into the chart



The chart is interactive. Move your mouse over it, and a small toolbar appears in the top-right corner. Click and drag directly on the chart to draw a box around any area - the chart will zoom in on exactly that spot, which is very useful for spotting small differences between the two lines. To zoom back out, click the little house icon in that same toolbar.

That's it - that's the whole tool. If anything still isn't clear, check the Documentation and Contact sections at the bottom of the DePeak page.